# EXPERIMENT 10: PERFORM MOVING OF AN IMAGE FROM ONE PLACE TO ANOTHER

## Aim

To perform image translation by moving an image from one position to another using Python and OpenCV.

## Algorithm

1. Import the OpenCV and NumPy libraries.

2. Read the input image using cv2.imread().

3. Get the height and width of the image.

4. Define the translation values Tx and Ty.

5. Create the translation matrix M = [[1,0,Tx],[0,1,Ty]].

6. Apply cv2.warpAffine() to move the image.

7. Display the original and translated images.

8. Save the translated image and terminate the program.

## Python Code

```
import cv2
import numpy as np

# Read input image
img = cv2.imread("input.jpg")

# Check image
if img is None:
    print("Image not found")
    exit()

# Get image size
rows, cols = img.shape[:2]

# Translation values
Tx = 100
```

```
Ty = 40

# Translation matrix
M = np.float32([[1, 0, Tx],
                [0, 1, Ty]])

# Move image
translated = cv2.warpAffine(img, M, (cols + Tx, rows + Ty))

# Save output image
cv2.imwrite("translated_image.jpg", translated)

# Display images
cv2.imshow("Input Image", img)
cv2.imshow("Translated Image", translated)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

## Input

## Output

## Result

Thus, the image was successfully moved (translated) from one position to another using affine transformation in OpenCV.